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Health Data Analysis Portfolio: Global Healthcare Accessibility & Outcomes

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Portfolio Overview:

This portfolio showcases my applied data analysis skills through a structured research project focused on global healthcare accessibility. It includes dataset selection, data cleaning, descriptive and inferential statistical analysis, and data visualization to evaluate relationships between healthcare access and key population health outcomes.

Tools & Skills Applied

- Excel (descriptive statistics, inferential statistics, z-scores)
- Data Cleaning & Visualization
- Correlation Analysis (Pearson r)
- Global health data interpretation
- Comparative analysis across income groups

Abstract

This study investigates the relationship between national healthcare access and health outcomes, concentrating on disparities among high- and low-income nations. Utilizing international datasets from the WHO, World Bank, UNICEF, CDC, and OECD, the research hypothesizes that improved healthcare accessibility positively correlates with better results, like life expectancy, maternal and infant mortality, and vaccination rates. To demonstrate this correlation, descriptive and inferential statistics were applied to global health data. Outcomes reveal that nations with wider healthcare access routinely have improved health outcomes, confirming the positive impact of equitable access and funding.

Introduction

Accessible healthcare is a vital aspect of population health as it affects fatality and disease rates. This study engages the hypothesis that a statistically significant, positive correlation occurs among national healthcare accessibility and essential health markers. The research and literature indicate a consistent gap in health outcomes among high- and low-income nations stemming from variations in healthcare institutional frameworks, funding, and accessibility. Data from recognized international organizations highlights the gravity of acknowledging these inequities through structural reform and international collaboration.

Methodology

This study uses supporting data from publicly open global databases:

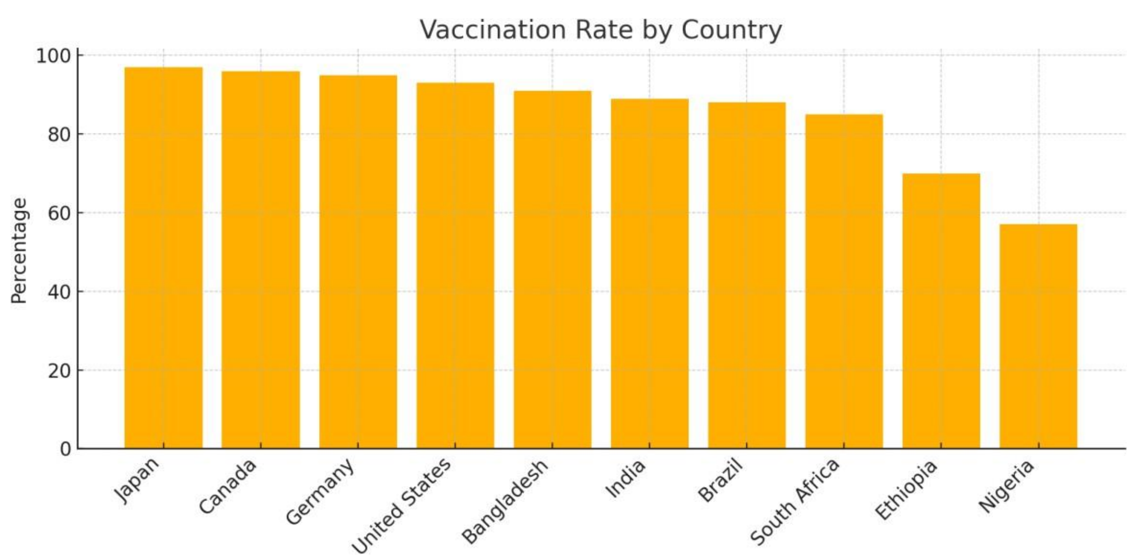
- World Health Organization (WHO) for life expectancy statistics.
- World Bank for health expenditure and maternal/infant mortality rates.
- UNICEF for global vaccination coverage.
- OECD for health spending in high-income countries.
- CDC for data on global health collaborations.

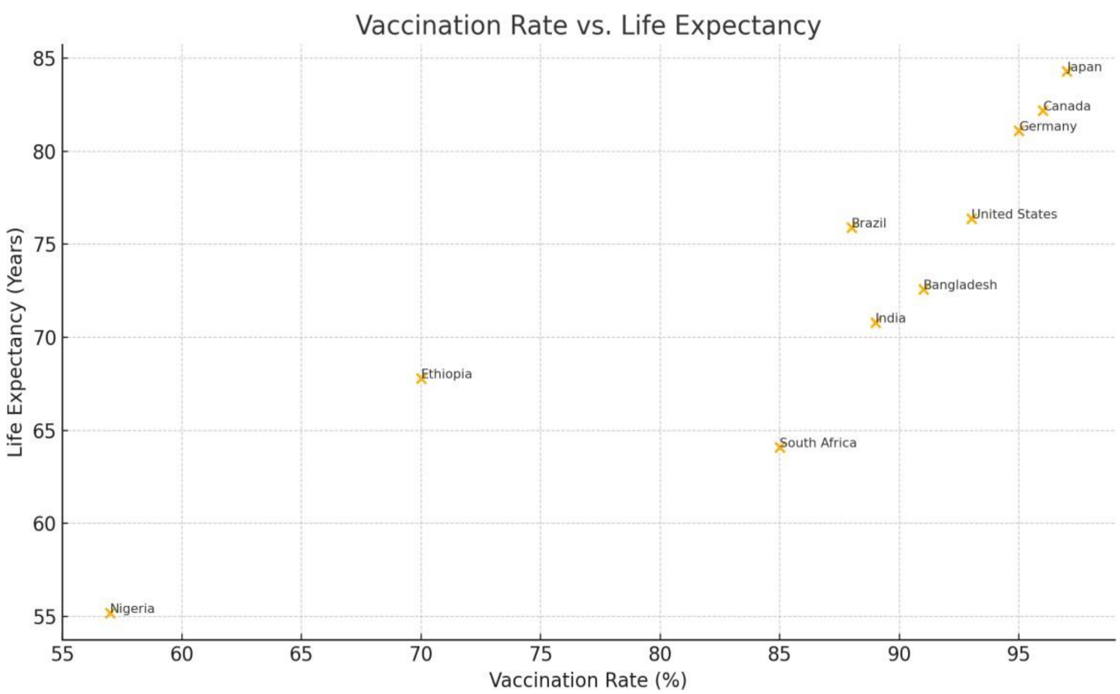
Descriptive statistics (mean, median, range) and inferential statistics (correlation coefficients, standardized z-scores) were assessed utilizing Excel. Data from at least 30 nations with varying income and economic demographics were examined to observe trends and correlations.

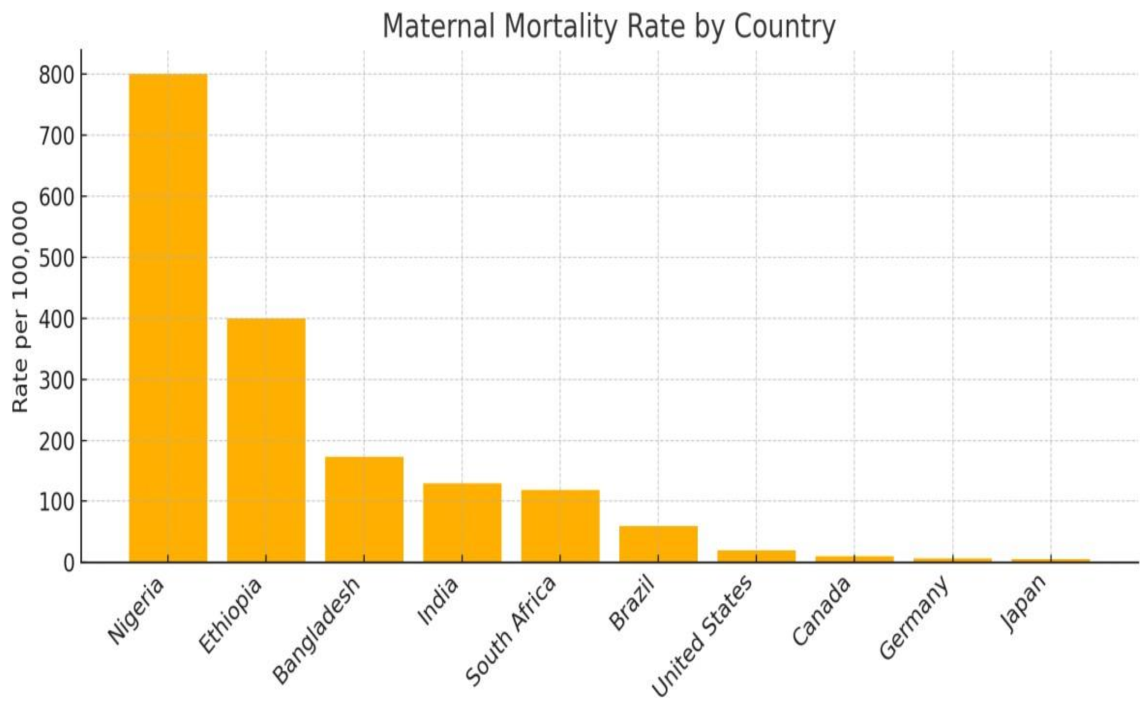
Analysis & Results

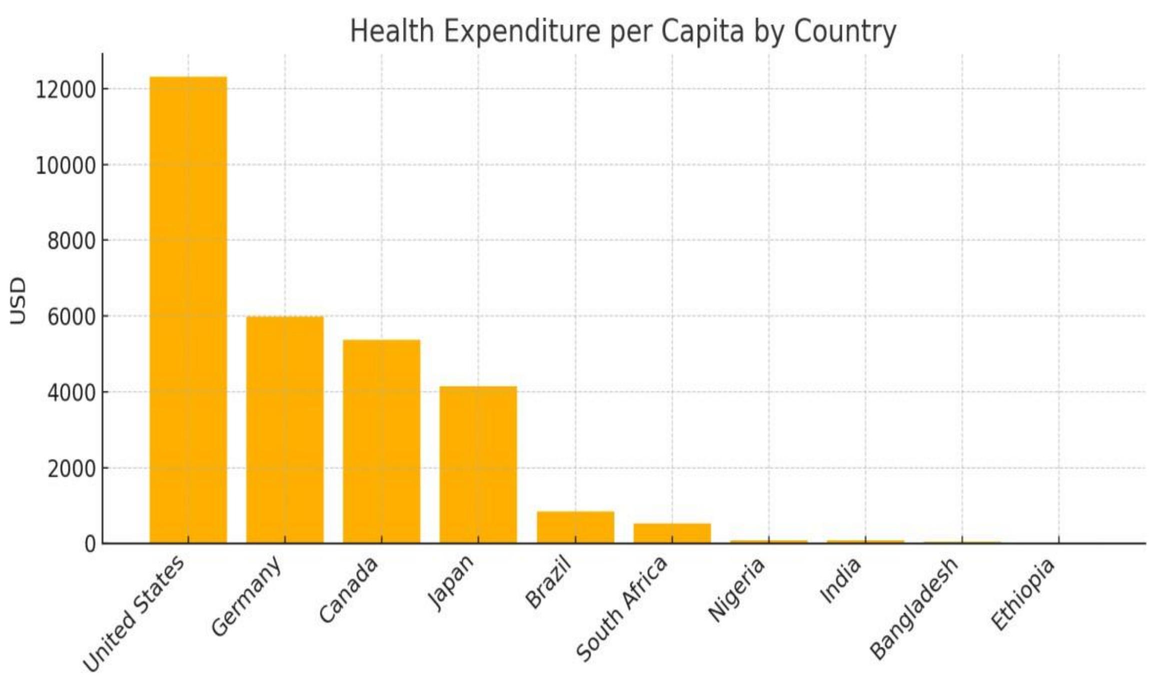
Descriptive statistics highlighted a clear variance in expected life span among countries with a high socioeconomic background (average: 81 years) and economically challenged nations (average: 64 years). Vaccination was over 90% in high-income countries but fell under 60% in various low-income countries. Maternal death rates were 10 per 100,000 live births in high-income nations in comparison to more than 400 observed in various low-income nations.

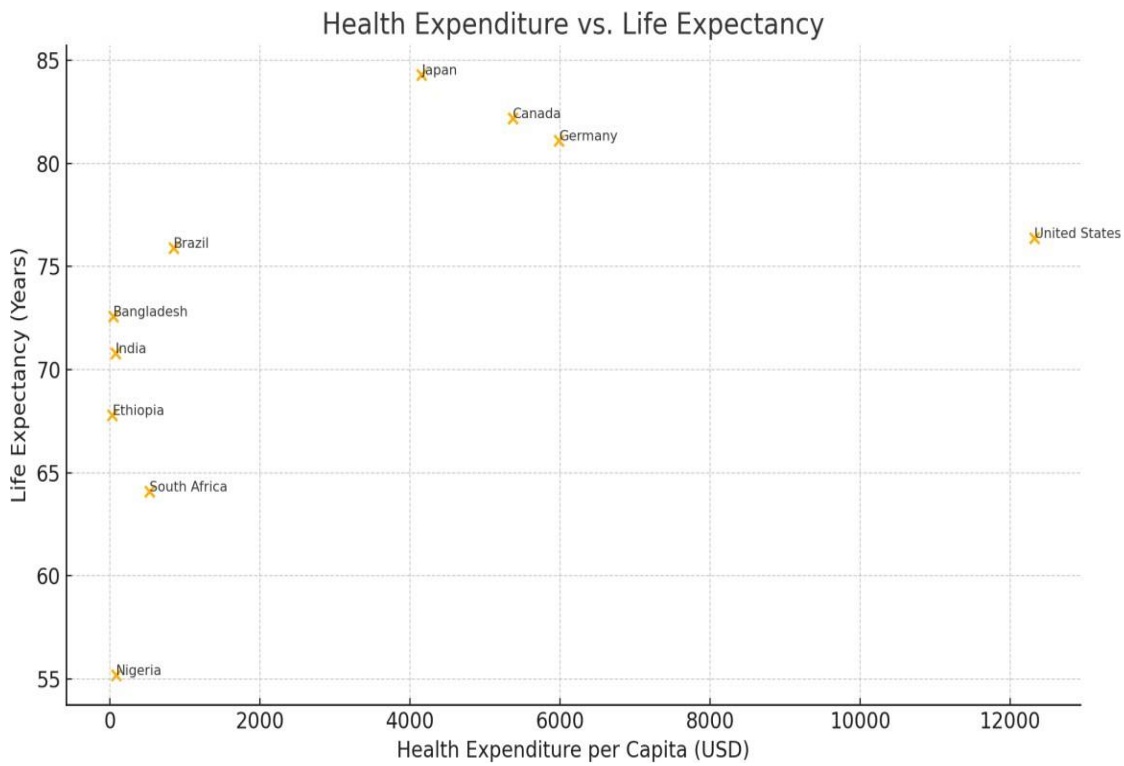
A Pearson correlation analysis identified a significant positive relationship between healthcare expenditure per capita and life expectancy ($r = 0.82$, $p < 0.001$), along with and a significant negative correlation between accessible healthcare and maternal mortality ($r = -0.76$, $p < 0.001$). The visual observable data was represented by scatterplots and bar charts illustrating these trends.

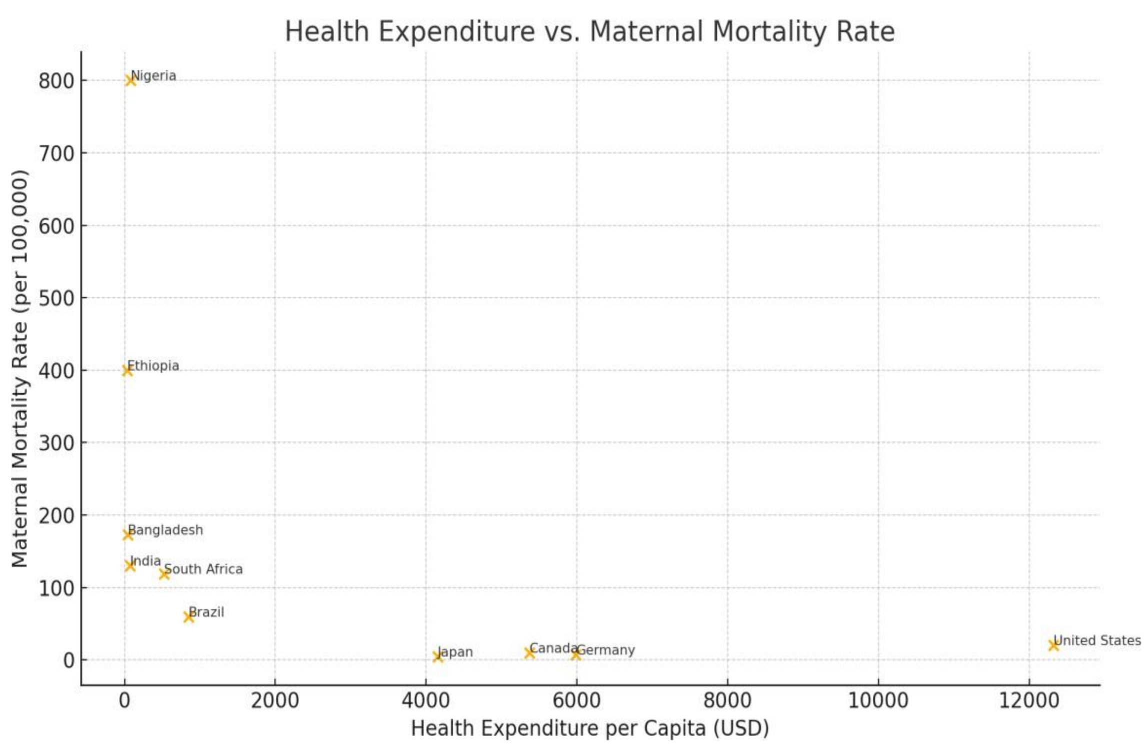



Vaccination Rate vs. Life Expectancy
Life Expectancy (Years)
Vaccination Rate (%)
85
80
75
70
65
60
55
55
60
65
70
75
80
85
90
95
Japan
Canada
Germany
United States
Brazil
Bangladesh
India
Ethiopia
South Africa
Nigeria


Maternal Mortality Rate by Country
Rate per 100,000
800
700
600
500
400
300
200
100
0
Nigeria
Ethiopia
Bangladesh
India
South Africa
Brazil
United States
Canada
Germany
Japan


Health Expenditure per Capita by Country
USD
12000
10000
8000
6000
4000
2000
0
United States
Germany
Canada
Japan
Brazil
South Africa
Nigeria
India
Bangladesh
Ethiopia


Health Expenditure vs. Life Expectancy
Life Expectancy (Years)
Health Expenditure per Capita (USD)
85
80
75
70
65
60
55
0
2000
4000
6000
8000
10000
12000
Japan
Canada
Germany
United States
Brazil
Bangladesh
India
Ethiopia
South Africa
Nigeria


Health Expenditure vs. Maternal Mortality Rate
Maternal Mortality Rate (per 100,000)
Health Expenditure per Capita (USD)
800
700
600
500
400
300
200
100
0
0
2000
4000
6000
8000
10000
12000
Nigeria
Ethiopia
Bangladesh
India
South Africa
Brazil
Japan
Canada
Germany
United States

Discussion

The findings confirm the hypothesis that greater accessible healthcare substantially impacts national public health outcomes. The distinct inequities among varying income demographics demonstrate the challenges that under-resourced healthcare institutions face, such as poorly funded health systems and underdeveloped infrastructure in disadvantaged areas. These results are consistent with previous studies referenced in the literature review. While wealthier countries gain an advantage from regular financial funding, poorer nations struggle to prevent public health emergencies as a result of limited resources. Constraints include the dependence on summarized national statistics, which can obscure regional inequities. Moving forward, the research should explore longitudinal analysis and integrate qualitative indicators to generate a better understanding.

References

Centers for Disease Control and Prevention. (n.d.). Global health. U.S. Department of Health & Human Services. https://www.cdc.gov/globalhealth/

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UNICEF. (2024). The State of the World's Children 2024 – Statistical compendium. https://www.unicef.org/reports

World Bank. (2024). World development indicators. https://databank.worldbank.org/source/world-development-indicators

World Health Organization. (2023). Global Health Observatory data repository. https://www.who.int/data/gho